

Summer Term 2026 – Julián Méndez & Ricardo Langner

# Interactive Information Visualization

Exercise: **Assignment 4**

# Introduction Assignment 4

## Dataset

We will use the same dataset as the last assignment, but with a focus on network vis and lenses. As before, the data is in the template.

The dataset consists of:

- 95 soccer players (nodes), for each player up to 40 data attributes
- ~1050 relations (edges) between players, an edge indicates if and how often the players have played for the same club during their career (edge weight, val)

# Introduction Assignment 3

## Programming Template

- Besides JavaScript and D3.js, we will use Cytoscape.js (<https://js.cytoscape.org/>).
- Cytoscape is specialized for networks. Look at examples with D3.js and compare!
- We will also learn one way to realize **semantic zooming** and **lenses**!
- Our template is based on [this example](#).

Template: <https://github.com/menocsk27/infovis-a4>

# Multivariate Visualization in Multiple Coordinated Views

## Semantic and Constant Density Zoom

Cytoscape.js already supports geometric zooming by mouse wheel or touchpad (feel free to adjust sensitivity). The template shows how to get the current zoom factor: `cy.zoom()`

- At level 0: only show nodes with *at least 10 attributes*, all with the same visual style.
- At level 1: show both nodes with less and more than 10 attributes, but different visual styles (e.g., circles versus squares).
- At level 2, show a D3.js star plot / radar chart on the nodes, with all available attributes.

We provide a function to draw star plots in the template, from [this source](#).

# Multivariate Visualization in Multiple Coordinated Views

## Semantic and Constant Density Zoom

Semantic level 2 can be computationally expensive, so the template provides a way to check if the nodes are visible in the current viewport.

Use this function to limit the number of star plots that are created!

# Multivariate Visualization in Multiple Coordinated Views

## Interactive Magic Lenses

You will implement a rudimentary version of a few magic lenses.

*The template reacts to mouse input/movement already (console output) and it already shows a circle for the lens.*

- Adapt the template so that the circle (your lens) follows the cursor.
  - Use the position of the mouse cursor (hover) to move the lens.
- Implement at least two lens functionalities:
  - *Node change* (alteration): show the star plots from the semantic zoom level 2 for the nodes inside the lens)
  - *Edge highlight*: visually highlight edges connected to nodes inside the lens

# When submitting your solutions:

## Interactive Magic Lenses

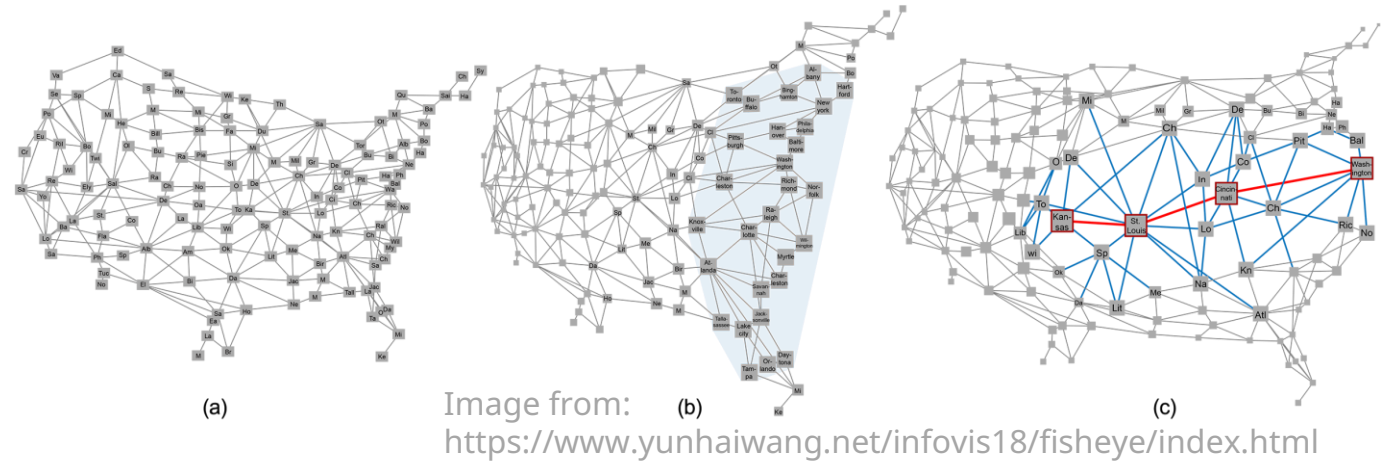
Add user interface controls for:

- Adjusting the size of the lens, e.g., a slider for setting the radius
- Enabling/disabling semantic zoom, e.g., a check box
- Switch between the two lens functionalities, e.g., a HTML select element

**Reflection:** Why is the edge highlight functionality (from the previous task) not the same as a [local edge lens](#)? Carefully think about this and give two arguments.

# When submitting your solutions:

## Extra task: Fisheye lens



Not enough? :) then keep going and implement a distortion lens.

The template includes d3-fisheye, which you can use if you decide to take on this challenge.

Exercise Interactive Information Visualization

# Thank you!

## Any questions?

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